

This document outlines the development of a simple Java-based login and registration system for the Programming POE. The system allows a single user to register and log in through a GUI, with features like password complexity validation, input fields, and authentication using a 2D array. Built with Apache Swing and following object-oriented principles, the project demonstrates the practical use of Java, form validation, and interface design in simulating a basic

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Programming POE Part I

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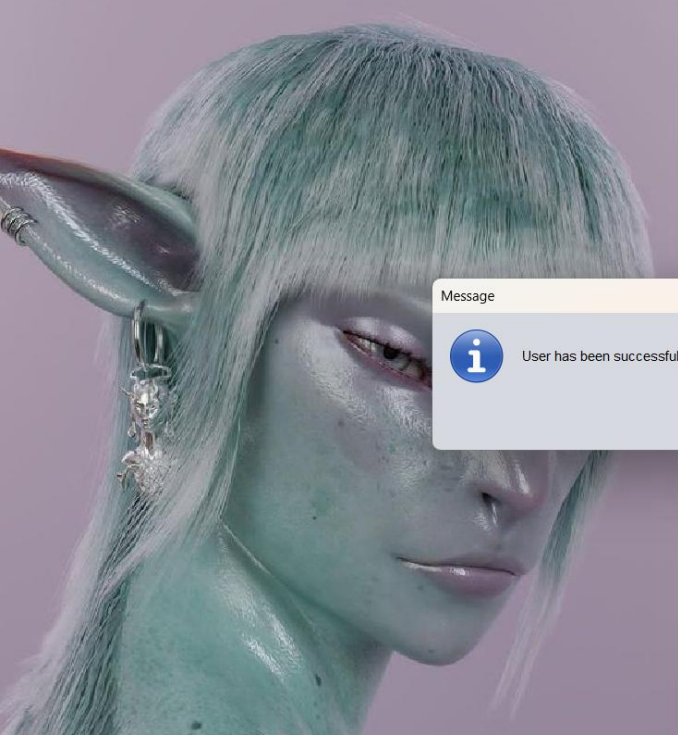
Introduction

This POE project's goal is to use Java to develop and implement a basic user authentication system. The two main features of this system are login and user registration. The program doesn't link to an external database and is made for a single user. Rather, during runtime, it stores the user's credentials in a two-dimensional array.

Using Apache Swing components, a graphical user interface (GUI) was created to improve the application's usability and facilitate user-friendly interaction. A key component of this system is password validation, which mandates that a password must adhere to complexity standards such a minimum length, the presence of a minimum of one capital letter, a number, and a special character and username validation which ensures the username consists of an underscore and is no more the 5 characters.

In order to facilitate readability and maintainability, the project uses a modular structure with distinct classes for handling functionality (such validation) and the user interface. This documentation describes the system's functionality, design choices, and development process.

GUI Outline



Sign Up

USERNAME
kyl_1

FIRSTNAME
Kyle

LASTNAME
RD

SECURITY
Ch&&sec@ke99!

Sign up

Message

 User has been successfully registered.

OK



Login

USERNAME

Login

Message

 Welcome Kyle, Zou it is great to see you

OK

SEE BEYOND THE SURFACE

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